Mingtian Tan

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Education

University of Michigan, Ann Arbor **Aug. 2024 – May 2026**
M.S. in Electrical and Computer Engineering (Robotics), GPA 3.5/4.0 *Ann Arbor, MI, USA*

- Coursework: Mathematics for Robotics, Robotic Systems Laboratory, Self-Driving Cars, 3D Robot Perception, Machine Learning, Robot Kinematics and Dynamics, Design Optimization

Xi'an Jiaotong-Liverpool University **Sep. 2020 – Jun. 2024**
B.Eng. in Electronic Science and Technology (dual degree, University of Liverpool), GPA 3.594/4.0 *Suzhou, China*

Publications

[1] **M. Tan** and V. Tzoumas, "Dual Gradient Consensus for Adaptive Learning to Transient Out-of-Distribution Disturbances in Quadrotor Control," under review at *IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2027.

[2] H. Zhou, **M. Tan**, and V. Tzoumas, "Efficient Domain-Adaptive Policy Learning via Kernel Representation with Application to Quadrotor Control under Non-Stationary Disturbances," arXiv:2606.13842 [cs.RO], 2026; under review at *IEEE Robotics and Automation Letters (RA-L)*.

[3] **M. Tan**, B. Gu, C. Gu, Y. Lu, M. Leach, Q. Zhang, and Q. Bu, "A Residential Robot for Chinese Cupping Physiotherapy," in *Robot Intelligence Technology and Applications 8 (RiTA 2023)*, Lecture Notes in Networks and Systems, vol. 1132, Springer, 2024, pp. 321–335 (oral presentation). doi:10.1007/978-3-031-70684-4_27

Research Experience

Intelligent Robotics and Autonomy Lab (iRaL), University of Michigan **Feb. 2025 – Present**
Research Assistant · Advisor: Prof. Vasileios Tzoumas, Department of Aerospace Engineering *Ann Arbor, MI*

Aerial manipulation with a flight-conditioned arm VLA (ongoing) *Oct. 2026 – Present*

- Designing a two-policy architecture for a fully actuated hexarotor with a 4-DoF arm in AM-Bench: a lightweight *Flight DiT* outputs flight action chunks and a flight-intent token that is injected into a $\pi_{0.5}$ arm VLA, which is then fine-tuned to fuse flight and arm control (project page).
- Built the demonstration dataset from scripted AM-Bench specialists instead of teleoperation: random drift on each specialist's target adds diversity while the labels keep the noise-free target, and only successful episodes are kept.
- Audited the official specialists and found their demonstrations unreliable (44/60 runs succeeded); redesigned five task policies, fixed a grasp-length error in the base environment, and added a seated-frame success check and a door-handle latch (60/60), then retuned the specialists for wind.
- Benchmarking the proposed Flight-DiT-conditioned $\pi_{0.5}$ against Diffusion Policy, LoRA-adapted $\pi_{0.5}$, and fully fine-tuned $\pi_{0.5}$ across 12 AM-Bench tasks and 48 disturbance variants (saturation, ground/wall effect, drag, wind).

Dual Gradient Consensus (DGC) for transient out-of-distribution disturbances *Jun. 2026 – Sep. 2026*

- Proposed DGC, a consensus-driven step-size rule for online residual learning: long-/short-term gradient memories and a cosine-agreement test sort each update into four regimes, a confidence map scales the step, and a hand-off rule switches domains only after persistent agreement [1].
- Built the full pipeline on top of DAPL (RFF residual predictor with low-pass pre-processing, offline BPTT policy, online least-squares adaptation) and the $\mathcal{L}_1$-augmented MPC, RFF-MPC, constant-step DAPL, and frozen baselines.
- Designed four mixed-disturbance benchmarks (step, stationary, periodic, and ramp shifts plus 0.1 s out-of-domain pulses), ten rollouts each; lowest mean position RMSE in every case (e.g., 0.145 vs. 0.255 m for DAPL).

Domain-adaptive policy learning via kernel representation (DAPL) *Feb. 2026 – Jun. 2026*

- Developed DAPL's learning pipeline [2]: the policy is trained offline by BPTT through differentiable simulation, and the RFF disturbance model is updated online by gradient descent.
- Built the benchmark harness (standardized reference trajectories and initial conditions, automated batch runs) and DATT (L1/RMA), $\mathcal{L}_1$-MPC, RFF-MPC, MPPI, and PPO baselines; ran the five-disturbance benchmark and σ/α ablations and produced the paper's RMSE tables and figures (up to 60–70% lower tracking error than baselines).
- Contributed reference-trajectory generation and control-loop timing fixes to the ROS 2 Crazyflie hardware stack.

Pathos: LiDAR-based agile quadrotor research platform *Sep. 2025 – Dec. 2025*

- Designed and built Pathos, a LiDAR-equipped agile quadrotor on the Agilicious stack: CAD and 3D-printed sensor payload and mounts, power/signal wiring under tight weight and space limits, and flight-tested vibration isolation.
- Integrated the onboard system on a Jetson Orin (ROS 1): Livox MID360 + FAST-LIO odometry, EKF pose fusion, a low-latency flight-controller–computer link, and monitoring/safety tooling; wrote the FAST-LIO guide.
- Unified localization, control, and planning into one real-time stack and deployed an RFF-based simultaneous system-identification MPC (SSI-MPC) controller, tuned on hardware; flight tests showed better disturbance rejection than baselines.

FAST Lab, Zhejiang University **May 2025 – Jul. 2025**

Research Intern · Advisor: Prof. Fei Gao, College of Control Science and Engineering *Huzhou, China*

- Extended the Swept-Volume Signed Distance Field (SVSDF) planner from offline to online swept-volume computation in dynamic environments.
- Devised a rigid-body-motion-based centroid–boundary formulation for continuous-time contact boundaries of arbitrary geometries, with exact constraints derived on the Lie group of rigid motions.
- Rewrote the swept-volume kernel with parallel CUDA primitives and coalesced memory access: 10× speed-up (40 ms → 4 ms per replan), enabling real-time obstacle-aware replanning.

Selected Projects

Autonomous Aerial Robot – B.Eng. Final-Year Project, XJTLU **Oct. 2023 – May 2024**

- Built and calibrated a quadrotor's power and perception stack; fused stereo VINS-Fusion visual-inertial odometry with high-rate flight-controller IMU data on a ROS 2 onboard computer for real-time pose estimation.
- Deployed and tuned the B-spline EGO-Planner for onboard planning and mapping and added state-machine safety logic for autonomous take-off and navigation; validated in outdoor waypoint missions (video).

Jiangsu Provincial Intelligent Robot Innovation Competition – Second Place **Aug. 2023 – Dec. 2023**

Team leader; entry: Residential Robot for Chinese Cupping Physiotherapy *XJTLU, Suzhou, China*

- Led the team from concept to live demo: adapted an ABB arm for home use with an STM32 + Raspberry Pi 4 dual-controller design and built a lightweight gripper with magnetic-encoder closed-loop rotation and air suction.
- Modeled the arm kinematics and planned the therapeutic trajectories, validated in controlled human testing; the entry was later published at RiTA 2023 [3] (video).

Technical Skills

Programming & ML	Python, C++, C, CUDA, MATLAB/Simulink; JAX, PyTorch; AI coding agents (Claude Code)
Learning & Control	MPC, adaptive and residual learning for control, RFF kernel residual models, differentiable simulation / BPTT policy optimization, diffusion policies, VLA fine-tuning
Robotics & Simulation	ROS 1/2, Agilicious, FAST-LIO, VINS-Fusion, EGO-Planner, Crazyflie/CrazySim, Isaac Lab/Isaac Sim, AM-Bench, LeRobot, OpenPI ($\pi_{0.5}$), Gazebo/URDF
Hardware & Prototyping	Quadrotor platform design and debugging; Jetson Orin, STM32, Raspberry Pi; Livox MID360 LiDAR; SolidWorks, 3D printing, laser cutting, soldering, vibration isolation

Honors and Awards

2023	Second Place, Jiangsu Provincial China University Intelligent Robot Innovation Competition
2022	Honorable Mention, Mathematical Contest in Modeling (MCM), Problem B
2020–2022	University Academic Achievement Scholarship (top 5%), XJTLU, 2020–21 and 2021–22

Professional Service

Reviewer	IEEE Int. Conf. on Robotics and Automation (ICRA) 2027